

Exercise – Understanding maximum simulated likelihood estimation

1. Aim of the exercise

This exercise estimates a random-coefficient model using maximum simulated likelihood (MSL). Whereas standard mixed-logit applications are typically reduced-form, the specification here is grounded in an explicit life-cycle economic model, demonstrating how random-coefficient methods can be applied within a more flexible structural framework.

2. Data

The life-cycle hypothesis predicts that older workers will gradually transition out of full-time employment by reducing their working hours prior to full retirement, yet participation in partial retirement remains low. Because few individuals actually participate in partial retirement, revealed-preference data provide limited scope for studying such behavior. Kantarcı et al. (2025) address this challenge by eliciting retirement preferences through a stated-choice survey. The survey was fielded in 2017 in the Longitudinal Internet Studies for the Social Sciences (LISS) panel and consists of two main components. The first collects background characteristics and information on respondents' work lives. The second is designed to elicit preferences for abrupt versus partial retirement. In this part, respondents answer a series of stated-choice questions in which they trade off working more hours in exchange for a higher pension against working fewer hours with a lower pension.

Figure 1 shows an example stated-choice question. It starts with a short introduction and then briefly describes three alternative retirement scenarios, followed by a timeline giving the number of hours worked and the earnings and pension income at each age. Respondents are asked to choose their favorite retirement scenario among the three, based on their own preferences.

To increase variation in choices, the question is asked to a given respondent for a total of three times, changing the ages of partial and full retirement in each case. The pension income is adjusted for working more or fewer years. The three versions use age 65, age 63 and 61 as the age of early or partial retirement, and age 70, age 68 and 66 as the age of late retirement, respectively. For ease of reference, we denote the three question variants – corresponding to early/partial retirement at ages 61, 63, and 65 – as retirement-age regimes 61, 63, and 65.

The duration of partial retirement is randomized between four and five years across two sub-samples of respondents. The survey also includes additional stated-choice questions. For the present exercise, the sample is restricted to respondents assigned to the five-year partial-retirement scenario, and the supplementary questions are excluded. These restrictions are made to simplify the exercise and they do not alter the core estimation problem.

3. Random utility model

To model the stated retirement preferences, Kantarcı et al. (2025) utilize a life-cycle framework. Assume that the total utility, U_i^q , of retirement trajectory q for individual $i = 1, \dots, n$ has the following form:

$$U_i^q = \sum_{t=60}^{100} \rho^{(t-60)} \pi_t U_{it}^q \quad (1)$$

where ρ is the discount factor. π_t is the probability of surviving from one age to the next, given

Many employees retire fully after working full-time; the age they retire can differ. Other employees go into partial retirement, where they work part-time for several years before full retirement.

Below we describe the retirement plans of three employees. All employees are 64 years old, work 40 hours a week, and earn €2,000 a month. Their retirement plans differ in the following respects:

- Age of retirement
- Pension income
- Type of retirement (partial or full retirement)

Please compare the plans presented below.

Amy plans to continue to work the same number of hours in the same job from age 65 to 69. She will retire at age 70. Her pension income will be €2,200 a month. This plan can be summarized as follows:

Age	62	63	64	65	66	67	68	69	70	71	72
	Work			Work					Retirement		
Hours worked	40 hours			40 hours					0		
Work income	€ 2,000			€ 2,000					0		
Pension income	0			0					€ 2,200		

Mary plans to reduce her hours to 20 hours a week and continue in the same job from age 65 to 69. She will earn €1,000 a month, and receive a partial pension income of €700 a month. While working part time, she will continue to build pension benefits for full retirement. She will retire fully at age 70. Her pension income will be €1,800 a month. This plan can be summarized as follows:

Age	62	63	64	65	66	67	68	69	70	71	72
	Work			Partial retirement					Retirement		
Hours worked	40 hours			20 hours					0		
Work income	€ 2,000			€ 1,000					0		
Pension income	0			€ 700					€ 1,800		

Linda plans to retire at age 65. Her pension income will be €1,400 a month. This plan can be summarized as follows:

Age	62	63	64	65	66	67	68	69	70	71	72
	Work			Retirement							
Hours worked	40 hours			0							
Work income	€ 2,000			0							
Pension income	0			€ 1,400							

Which plan do you find the most attractive?

- ☐ Amy's plan
- ☐ Mary's plan
- ☐ Linda's plan

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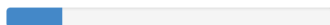


Figure 1: The stated preference question asking to choose among early, partial and late retirement

survival up to age 60. U_{it}^q is the utility at age $t = 60, \dots, 100$. The time horizon is fixed at 100 years of age. q is an early abrupt retirement trajectory (E), a partial retirement trajectory (P), or a late abrupt retirement trajectory (L). In all trajectories, the agent is working full-time at age 60. At later ages t , leisure l_{it}^q and net income y_{it}^q vary across trajectories.

Within period utility is specified as follows:

$$U_{it}^q = \alpha_{it}^l \ln(l_{it}^q) + \alpha^y \ln(y_{it}^q) + \alpha^{ly} \ln(l_{it}^q) \ln(y_{it}^q) \quad (2)$$

$$\alpha_{it}^l = X_i \beta^l + \eta^l t + e_i^l \quad (3)$$

$$e_i^l \sim N(0, \sigma_l^2) \text{ and } e_i^l \text{ independent of } X_i \quad (4)$$

$$l_{it}^q = T - h_{it}^q \quad (5)$$

where T is the number of hours available for work and leisure in a working week and is a parameter to be estimated. h_{it}^q denotes hours of paid work per week. At each age t , the person can work full-time ($h_{it}^q = 40$), can be partially retired ($h_{it}^q = 20$), or can be fully retired ($h_{it}^q = 0$).

During full-time employment, net income y_{it}^q consists of after-tax earnings. During full retirement, it corresponds to after-tax pension income. During partial retirement, y_{it}^q includes both part-time earnings and partial pension income.

The preference parameters α_{it}^l and α^{ly} drive the marginal utility of leisure time for respondent i at age t . α_{it}^l depends on observed characteristics X_i such as age, gender and home ownership, and, through e_i^l , on respondent i 's unobserved characteristics. The effect of age t is captured by $\eta^l t$. We expect $\eta^l > 0$, since respondents' valuation of leisure will typically increase with age. A nonzero parameter α^{ly} implies that the marginal utility of leisure also varies with income.

Together with α^{ly} , the parameter α^y determines the marginal utility of income. Both parameters are treated as constants, to avoid multicollinearity and imprecise estimates.

Introducing error terms u_i^q as in a standard random utility model, the model takes the following form:

$$V_i^q = U_i^q + u_i^q \quad (6)$$

$$u_i^q \sim \text{i.i.d. type I extreme value and independent of } X_i, e_i^l \quad (7)$$

$$F(u_i^q) = e^{-e^{-u_i^q}} \quad (8)$$

where F denotes the cumulative distribution function.

The observed choice in question Q is given by

$$C_i^Q = q \text{ if } V_i^q > V_i^p \text{ for all } p \neq q. \quad (9)$$

As described earlier, as we consider in the present exercise, respondents choose among retirement trajectories in three questions.

Define $u_i^q - u_i^p \equiv u_i^{qp}$. The assumptions on u_i^q imply that u_i^{qp} has a standard logistic distribution and that the probability of choosing scenario q among alternative scenarios j in question Q , given all individual and scenario characteristics and preference parameters (including the unobserved preference term e_i^l), is given by

$$P(C_i^Q = q \mid A_i, e_i^l) = \frac{e^{U_i^q}}{\sum_j e^{U_i^j}} \quad (10)$$

where $A_i = \{l_{it}^q, y_{it}^q, X_i, t, \beta^l, \eta^l, \alpha^y, \alpha^{ly}\}$ is the set of all relevant individual and trajectory characteristics and parameters.

Model estimation is similar to estimation of a mixed logit or other random coefficient models as in, e.g., Revelt and Train (1998), using MSL. The (unconditional) likelihood contribution for individual i can be written as a one-dimensional integral over the unobserved heterogeneity e_i^l of the product of the conditional probabilities of the observed outcomes C_i^Q :

$$L_i = \int_{-\infty}^{\infty} \prod_{Q=1}^{K(i)} P\left(C_i^Q = q \mid A_i, e_i^l\right) \frac{1}{\sigma_l} \phi\left(\frac{e_i^l}{\sigma_l}\right) de_i^l \quad (11)$$

where $K(i)$ is the number of questions answered by respondent i , which is 3 in the current exercise.

Since it is not feasible to compute the integral numerically, we approximate it using simulated values of the random coefficient and use simulated maximum likelihood (?), replacing the integral with the following sum

$$\frac{1}{S} \sum_{s=1}^S \prod_{Q=1}^{K(i)} P\left(C_i^Q = q \mid A_i, e_{i,s}^l\right) \quad (12)$$

where S is the number of simulations and $e_{i,s}^l$ are random draws from a normal distribution with mean 0 and standard deviation σ_l . Usually a large number of pseudo-random draws is needed to assure a reasonably low simulation error in the estimated parameters. The number of draws and thus the time the estimation procedure takes can be reduced (keeping the same simulation variance) by using quasi-random numbers of Halton sequence. Estimates of the covariance matrix of the parameter estimates are based upon asymptotic results, e.g., in Gouriéroux and Monfort (1990).

5. Implementation

Here we explain the MSL estimation procedure. The procedure builds on econometric concepts covered in earlier exercises in Simulation Methods in Econometrics: Theory and Applications in MATLAB, including Monte Carlo integration and quasi-random sampling. Readers are therefore advised to work through those exercises before proceeding. The estimation procedure in MATLAB is organized into four script files and two function files. The first script file, `load_data.m`, loads the data. The second script file, `generate_halton_draws.m`, calls the first function file, `halton.m`, to generate quasi-random draws. The third script file, `likelihood_estimation.m`, calls the second function file, `likelihood.m`, to initiate the MSL estimation. The contents of each file are explained below.

5.1. Load the data

We begin by loading the data into the Workspace:

```
18 load('data.mat')
```

Once the data are loaded, the Workspace contains the following variables. `id` contains the identification numbers of respondents.

`X_i` is the matrix of observed characteristics, including variables such as age, gender, and home ownership.

$\mathbf{t}$ is a row vector with 41 values, ranging from ages 60 to 100. It indexes the age of the hypothetical individual and is used to construct total utility, as defined in equation (1).

The variable `choice_at_age_regime_61` records which of the three retirement scenarios a respondent selects in the question corresponding to retirement-age regime 61. A value of 1 indicates early (abrupt) retirement, 2 indicates partial retirement, and 3 indicates late (abrupt) retirement. The variables `choice_at_age_regime_63` and `choice_at_age_regime_65` contain the analogous choices for the questions featuring retirement-age regimes 63 and 65, respectively.

The variable `h_i_t_65E` contains, in each row, the number of hours worked per week at each age from 60 to 100 for a given respondent, for the retirement trajectory in which early retirement begins at age 65. This trajectory is illustrated at the bottom of Figure 1: the hypothetical individual works 40 hours per week from ages 60 to 64 and retires at age 65, after which weekly hours drop to zero. Accordingly, the first five columns of `h_i_t_65E` take the value 40 (for ages 60–64), and all remaining columns take the value 0. Other variables are defined analogously for the remaining retirement trajectories. For example, `h_i_t_61E` contains a value of 40 in the first column (age 60), after which weekly hours drop to zero. These variables are used to construct total utility.

The structure of the variable `y_i_t_65E` mirrors that of `l_i_t_65E`. It contains, in each row, the monthly income at each age from 60 to 100 for the retirement trajectory in which early retirement begins at age 65. As illustrated at the bottom of Figure 1, the hypothetical individual earns a monthly wage of €2,000 from ages 60 to 64 and retires at age 65, after which she receives a pension income of €1,400. Accordingly, the first five columns of `y_i_t_65E` take the value €2,000 (for ages 60–64), and all remaining columns take the value €1,400. Other variables are defined analogously for the remaining retirement trajectories.

5.2. Generate Halton draws

Next, we run the third script file, `generate_halton_draws.m`. At the end of this script, the function file `halton.m` is called as follows:

```

40 [~,Z] = halton(N_i,halton_N_dimensions,halton_N_draws_per_i,...
41     'prime',halton_primes,...
42     'burn',halton_N_burns,...
43     'leap',halton_N_leaps,...
44     'random',halton_randomize_indicator,...
45     'scramble',halton_scramble_indicator);

```

The function returns `Z`, a three-dimensional array of size $N_i \times \text{halton_N_dimensions} \times \text{halton_N_draws_per_i}$. It contains quasi-random draws transformed to standard normal. The first dimension corresponds to the number of respondents and assigns each individual a unique sequence of Halton points. The second dimension represents the number of prime bases used and therefore the dimensionality of the integral over the random coefficients. The third dimension specifies the number of quasi-random draws per respondent used to approximate the integral in the simulated likelihood.

5.3. Construct the likelihood

The function file `likelihood.m` constructs the objective function. We begin by specifying the function name, its input arguments, and the objective function as its output.

Next, we create the scalar array `N_k`, which specifies the number of covariates included in

the model. The parameter vector `theta` contains all parameters to be estimated. Its first `N_k` elements correspond to the covariate-specific coefficients collected in `beta_1`, while the remaining elements of `theta` are sequentially assigned to scalar parameters.

The array `e_i_1` represents the error term entering the random-coefficient equation. To generate these errors, we scale standard normal draws by the standard deviation parameter `sigma_1`. Specifically, we set

```
55 e_i_1(:,1,:) = sigma_1 * Z(:,1,:);
```

where `Z` contains standard normal draws. This operation produces normally distributed errors with variance `sigma_1` for each individual `i` and each Halton draw.

`e_i_1` inherits its dimensionality from the structure of the quasi-random draws `Z` as described above.

We define the random coefficient using `X_i`, `t`, and `e_i_1` as specified in equation (3):

```
55 alpha_i_t_1 = X_i * beta_1 + eta_1 * t + e_i_1;
```

Next, equation (1) requires defining the leisure and income terms in logarithmic form. For illustration, the leisure term for the early-retirement scenario at the retirement-age regime 65 is given by

```
68 log_T_h_i_t_65E = log(T-h_i_t_65E);
```

and the income term is defined analogously as

```
83 log_y_i_t_65E = log(y_i_t_65E);
```

Given the leisure and income terms defined above, we are now ready to specify the within-period utilities given by equation (2). For illustration, the within-period utility for the early-retirement scenario at the retirement-age regime 65 is given by

```
125 U_i_t_65E = alpha_i_t_1 .* log_T_h_i_t_65E ...
126           + alpha_y .* log_y_i_t_65E ...
127           + alpha_l_y .* log_T_h_i_t_65E .* log_y_i_t_65E;
```

The vector `rho_0_to_40` contains the discount factors for ages 60 to 100, with the exponent `(t-60)` ensuring that discounting begins at age 60. The vector `pi` stores the corresponding survival probabilities from age 60 onward. Together, these objects provide the discounted survival weights used in computing expected lifetime utility in the early-retirement scenario at the retirement-age regime (65), given by

```
165 U_i_65E = sum(rho_0_to_40 .* pi .* U_i_t_65E,2);
```

Next, we construct equation (10) as follows:

```
239 lik_choice_at_age_regime_65 = ...
240     sum(exp_U_i_65 .* ind_choice_at_age_regime_65,2) ./ ...
241     sum(exp_U_i_65,2);
```

This expression consider the retirement-age regime 65, and gives the probability of choosing a particular scenario by multiplying the indicator for that scenario with the exponentiated utility and then dividing by the total exponentiated utility across all scenarios.

We now define the joint probability of choosing scenarios at three questions for each `i`:

```

247 lik = lik_choice_at_age_regime_61 .* ...
248       lik_choice_at_age_regime_63 .* ...
249       lik_choice_at_age_regime_65;

```

`lik` is a three-dimensional array of size $N_i \times \text{halton_N_dimensions} \times \text{halton_N_draws_per_i}$, containing the joint probability of the three scenario choices for each respondent and each simulation draw. Recall that the Halton draws index the simulated random-coefficient errors, so the likelihood must be evaluated separately for each draw.

We average `lik` over the Halton draws for each `i`:

```

256 avg_lik = mean(lik,3);

```

Averaging over the third dimension collapses the draw-specific likelihoods. Since `lik` contains one likelihood value per individual, per random-coefficient dimension, and per simulation draw, this operation yields a $N_i \times 1$ vector representing the simulated likelihood for each respondent after integrating over the random-coefficient distribution.

The next two steps transform the simulated likelihoods into a form suitable for numerical optimization. First, we take logs:

```

262 log_avg_lik = log(avg_lik);

```

This converts the simulated likelihood for each individual into its log-likelihood contribution.

We then sum these log-likelihood contributions across individuals and multiply by -1 to obtain the negative log-likelihood, which is the objective function minimized during estimation:

```

268 sum_log_avg_lik = -sum(log_avg_lik);

```

5.4. Estimation

The script file, `likelihood_estimation.m`, calls the function file, `likelihood.m` as follows:

```

58 [theta_hat,fval,exitflag,output,~,grad,hessian] = ...
59 fmincon(obj,theta_ig,[],[],[],[],[],[],[],options);

```

Here, `obj` denotes the objective function, which is defined as the negative of the simulated log-likelihood evaluated at a given parameter vector. The vector `theta_ig` contains the initial guesses for the parameters to be estimated. These initial values serve as the starting point for the numerical optimization routine. The function `fmincon` then searches over the parameter space to find the value of `theta` that minimizes the objective function, thereby maximizing the simulated log-likelihood.

5.5. Results

The script file `results_table.m` compiles the parameter estimates, standard error estimates, and performs inference using the corresponding t -statistics and confidence intervals. Table 1 presents the results.

Table 1: Estimation results

Parameter	Estimate	Standard error	t value
β^l : constant	-7.013	0.836	-8.393
β^l : age	-0.759	0.171	-4.447
β^l : male	-0.197	0.038	-5.216
β^l : high education	-0.047	0.030	-1.586
β^l : household with no children	0.031	0.044	0.698
β^l : with partner	0.067	0.033	2.028
β^l : home owner	0.042	0.033	1.261
β^l : had a health problem in the last six months	0.053	0.030	1.773
β^l : would work even if money was not needed	-0.082	0.013	-6.467
β^l : experienced or expect early retirement	0.190	0.040	4.749
η^l	0.100	0.010	9.657
σ_l	0.425	0.047	9.056
T	50.191	1.537	32.650
α^y	-0.579	0.158	-3.659
α^{ly}	0.233	0.046	5.023
ρ	1.022	0.015	65.797

6. Final notes

This file is prepared and copyrighted by Hartog Horsch and Tunga Kantarcı. The file and the accompanying MATLAB code are publicly available on GitHub and can be accessed via the following link: [link](#). This exercise draws on two references. The first is: Kantarcı, T., Been, J., van Soest, A., van Vuuren, D. (2025). Partial retirement opportunities and the labor supply of older individuals. Labour Economics, 96, 102739. Available at: [link](#). The second is: Revelt, D., Train, K., 1998. Mixed logit with repeated choices: Households' choices of appliance efficiency level. The Review of Economics and Statistics 80 (4), 647-657. Available at: [link](#). The third is: Gouriéroux, C., Monfort, A., 1990. Simulation based inference in models with heterogeneity. Annals of Economics and Statistics, (20/21), 69-107. Available at: [link](#).